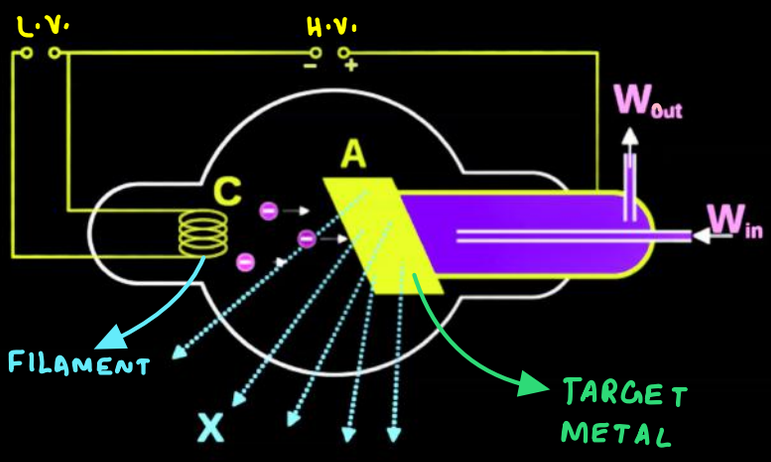
Production Of X-Rays Using A Coolidge Tube:

Production of x rays is also sometimes known as reverse photoelectric effect because here high energy electrons are incident on a target metal and the metal emits photons at the expense of electron energy. In a typical coolidge tube, electrons are emitted from a hot filament due to thermionic emission. These electrons are accelerated through very high voltages (Typically 5kV to 10^5 Volts). The accelerated electrons are the made to strike a target metal which is provided with cooling tubes to prevent its melting and excessive heating. Due to high energy collisions of electrons in the metal, two types of x-rays are produced about which we will study in the upcoming sections. X rays typically have wavelengths of the order of 1Å.

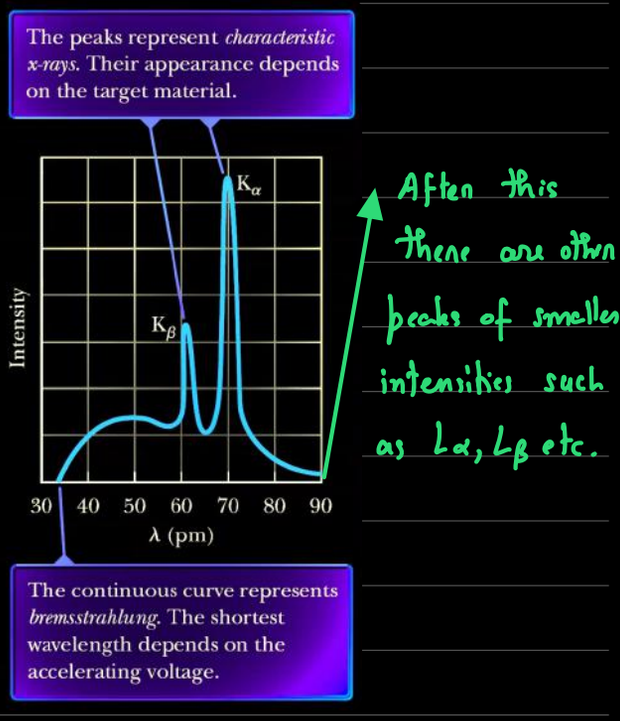


COOLIDGE TUBE

Continuous And Characteristic X-Rays:

When the accelerated electrons hit the target metal, they may either lose their energy in random collisions in the metal matrix, thus creating photons at the expense of the electron energies. This radiation is also known as Bremmstahlung (or braking radiation, because electrons are being "braked"). The photons emitted through this mechanism can have a maximum energy equal to the maximum energy of the electrons. Thus if V_0 is the accelerating potential, the emitted photons will have a "cut-off" or a minimum wavelength given by:

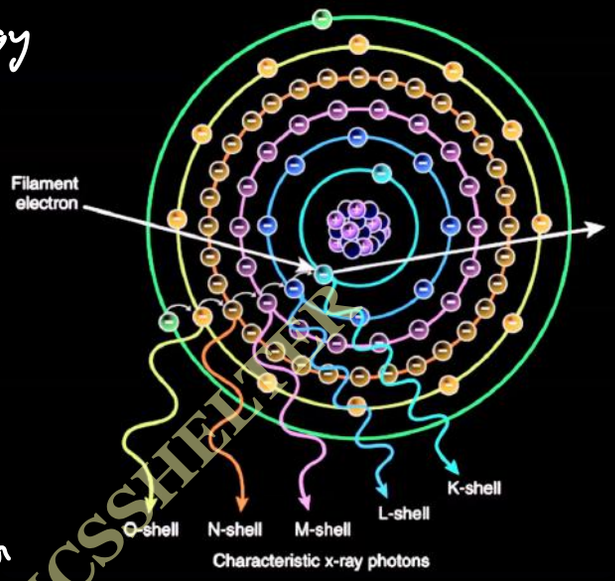
$eV_0 = hc/\lambda_0 \Rightarrow \boxed{\lambda_0 = hc/eV_0}$ ← cut-off or threshold wavelength.



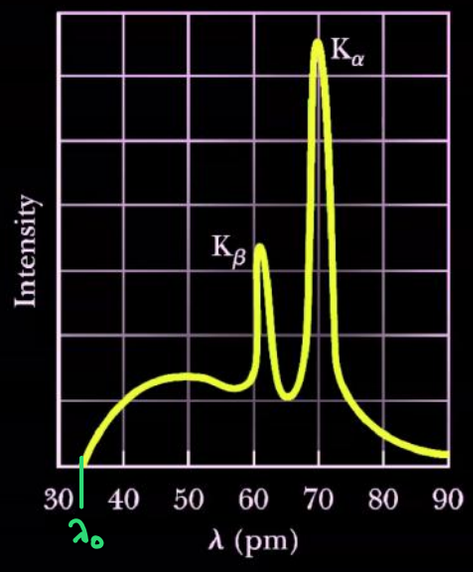
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CONTINUOUS X-RAYS: Since through random collisions, emitted photons can have any arbitrary wavelength ranging between λ_0 and ∞, the x-rays generated through this mechanism are called continuous x-rays.

CHARACTERISTIC X-RAYS: Some of the high energy electrons may be successful in ionizing the target metal atoms by knocking out an electron from one of the shells. In that case a vacancy is created and electrons from higher shells jump to fill in the vacancy and in this process photons are emitted just as in deexcitation for hydrogen like atoms. The emitted radiation in this manner depends on the energy levels of the target atom and hence known as characteristic radiation. Characteristic radiation is named based on the shell from which the electron has been knocked and the number of level jumps in the photon emission process, for instance K_α means electron was knocked from K shell and vacancy fulfilled by electron jump from L shell (α: 1 level jump), similarly K_β means a jump from M shell to K shell.



→ Note that K_β photons are more energetic than K_α photons but still K_α peak is higher because there are much larger number of K_α photons than K_β photons. (A vacancy in K shell is more likely to be filled by jump from L shell rather than that from M shell).

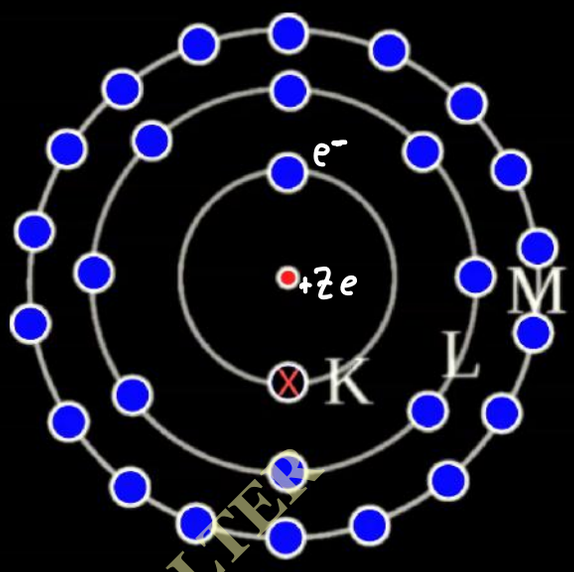


→ To observe K_α emission, minimum required electron energy corresponds to the transition $K \to \infty$ (corresponding terms K absorption edge (wavelength), K exciting potential (potential difference), Short wave cut off (wavelength/frequency).

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Understanding X-Rays In Terms Of Bohr Model:

If one electron from K shell is knocked out then to electrons in the L shell, the effective charge of the nucleus appears to be $(Z-1)e$. (For simplistic modeling we may assume the average position of the remaining electron of K shell as the centre of the orbit i.e the nucleus). Thus for a simplistic model, we may assume that an electron in L shell has to jump to K shell to emit photons in a manner similar to that in hydrogen like atoms. Thus:



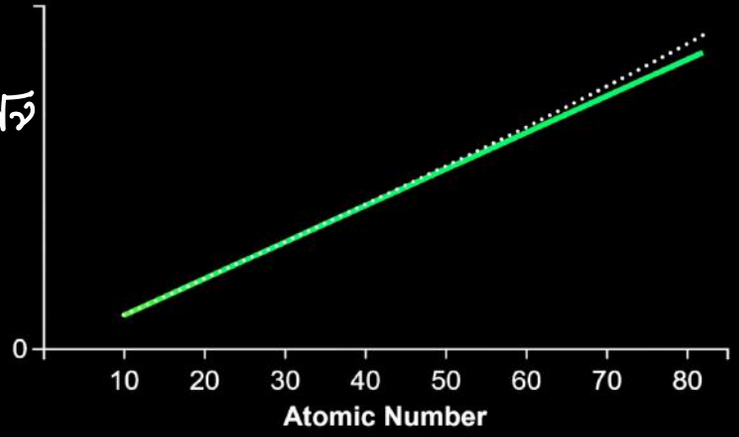
$$h\nu = Rhc(Z-1)^2\left[1-\frac{1}{4}\right] \quad (\because \Delta E = E_0 Z^2\left[\frac{1}{n^2}-\frac{1}{m^2}\right])$$

or $$\sqrt{\nu}_{K\alpha} \approx \sqrt{\frac{3Rc}{4}}\,(Z-1)$$

Thus using our simplistic model we expect that K_α frequency must be proportional to $(Z-1)$. Though our model has some limitation for example averaging an inverse square field was done too simplistically and repulsion from other electrons was ignored. For a more accurate analysis, these factors must also be accounted for. A similar analysis can also be done for L_α and M_α lines etc with properly taking shielding factors in to account.

Moseley's Law:

Moseley studied the variation of square roots of K_α frequencies of various elements with their atomic numbers and found that the variation was linear. Infact at that time the periodic table was arranged in order of masses and thus



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Cobalt (58.9) came after Nickel (58.7) and thus they did not fit the graph properly, however when swapped, the points lied accurately on the graph and this gave hints about atomic number i.e proton numbers were a fundamental property of elements just like their mass numbers. Mathematically, the moseley's law is expressed as:

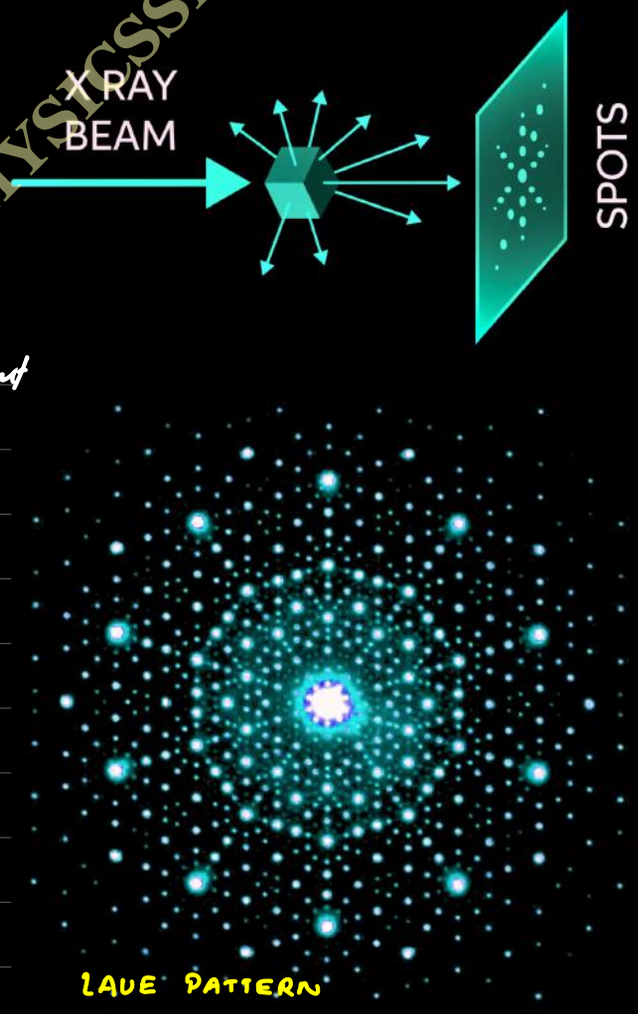
$$\sqrt{\nu} = a(Z-b)$$ Here a and b are suitable constants.

Alternatively, $$\nu = A(Z-b)^2$$ where $b=1$ for K_α and 7.4 for L_α

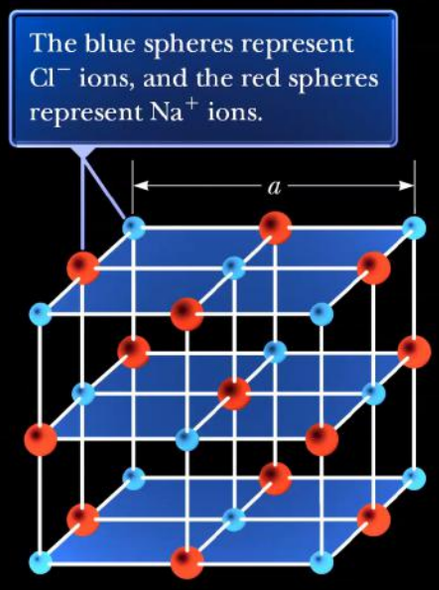
and $$A = 3Rc/4 \approx 2.5 \times 10^{15}\ Hz$$

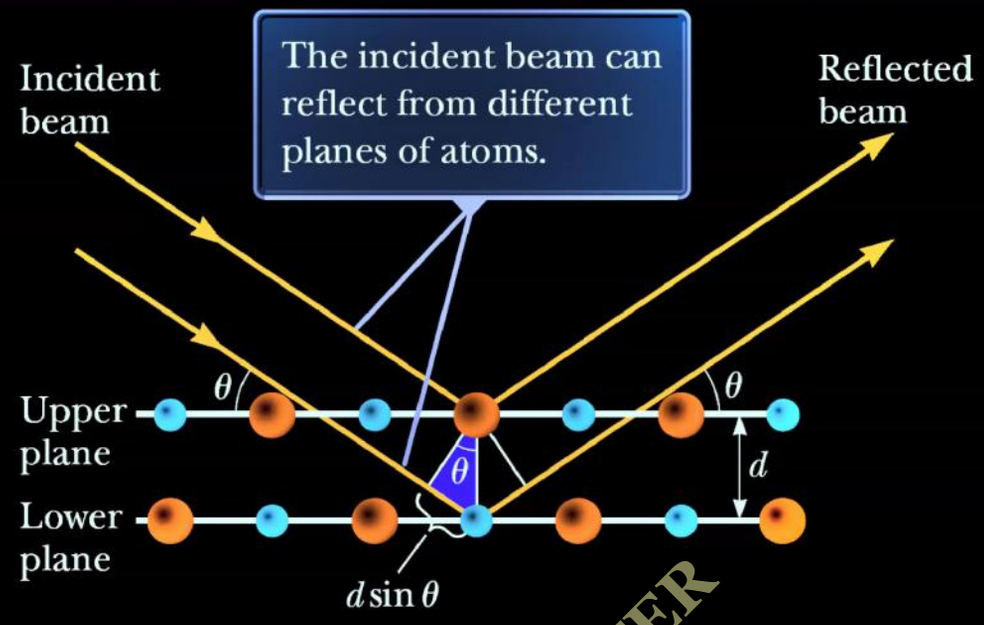
X-Ray Diffraction: [Bragg's Law):

Recall from wave optics how multiple slits (Diffraction grating) could form maxima and minima on a screen. In nature crystals serve as natural 3 Dimensional Diffraction gratings due to periodic arrangement of atoms in various planes. Since x-rays have wavelength of order of 1Å which is same as that of interatomic spacing in various crystals, x-ray diffraction can be used for figuring out spacing and arrangement of atoms in a crystal. Diffracted beams are very intense in certain directions due to constructive interference from waves reflected from various layers of crystals. The diffraction patterns are in general complex due to 3D nature of crystal structure. In the image on next page we can see a sodium chloride crystal in which we can see that atoms are arranged in various planes. Now if an x ray beam makes an angle θ with one of the planes, as shown in the following figure, the path difference between



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the rays reflected from upper and lower planes is $2d \sin\theta$. Thus for constructive interference, $\boxed{2d \sin\theta = n\lambda}$ ← BRAGG's LAW.

EXAMPLE: IRODOV 6.133:

Find the wavelength of the K_α line in copper ($Z = 29$) if the wavelength of the K_α line in iron ($Z = 26$) is known to be equal to 193 pm.

Solution: $\nu = A(Z-1)^2 = \frac{c}{\lambda} \Rightarrow \frac{\lambda_{Cu}}{\lambda_{Fe}} = \left(\frac{Z_{Fe}-1}{Z_{Cu}-1}\right)^2$

$\Rightarrow \lambda_{Cu} = 193\,\text{pm} \times \left(\frac{25}{28}\right)^2 \approx 154\,\text{pm}$ Ans.

EXAMPLE: IRODOV 6.134:

Proceeding from Moseley's law find:
(a) the wavelength of the K_α line in aluminium and cobalt:
(b) the difference in binding energies of K and L electrons in vanadium. Give $Z_{Al} = 13$, $Z_{Co} = 27$, $Z_V = 23$

Solution: Moseley's law is just like Bohr model with Z replaced by $Z-1$ and transition considered from $2 \to 1$. Thus:

$\lambda = \frac{\lambda_0}{(1-\frac{1}{4})(Z-1)^2}$, thus for Al, $\lambda_{Al} = \frac{911\,\text{Å}}{12^2} \times \frac{4}{3} = 8.435\,\text{Å}$

$= 843.5\,\text{pm}$ Ans.

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For L_α, $\lambda_0 = \dfrac{911\,A^\circ}{(1-1/4)(26)^2} \approx 180$ pm Ans.

(b) Binding energy for K shell : $h\nu_{K\alpha}$ similarly for L_{shell} it is $h\nu_{L\alpha}$, thus Difference: $h\nu_{K\alpha} - h\nu_{L\alpha} = h\nu_{K\alpha}$.

Thus the required difference can be obtained using hydrogen model:

$$\Delta E = 13.6\,eV \times (23-1)^2\left[1-\frac{1}{4}\right] \approx 5\,keV \text{ Ans.}$$

EXAMPLE: IRODOV: 6.135:

How many elements are there in a row between those whose wavelengths of K_α lines are equal to 250 and 179 pm?

Solution: $\lambda = \dfrac{\lambda_0}{(1-1/4)(z-1)^2} \Rightarrow z = \sqrt{\dfrac{4\lambda_0}{3\lambda}} + 1$

Thus for 250 pm, $z = \sqrt{\dfrac{91126\times 4}{3\times 250}} + 1 \approx 23$

For 179 pm, $z = \sqrt{\dfrac{91126\times 4}{3\times 179}} + 1 = 27$

Thus there are three elements in between i.e $z = 24, 25$ and 26. Ans.

EXAMPLE: IRODOV: 6.136:

Find the voltage applied to an X-ray tube with nickel anticathode if the wavelength difference between the K_α line and the short-wave cut-off of the continuous X-ray spectrum is equal to 84 pm. ($Z = 28$)

Solution: $\lambda_{K\alpha} = \dfrac{\lambda_0}{(3/4)(z-1)^2} = \dfrac{4\times 911\,A^\circ}{3\times 27^2} \approx 167$ pm, thus,

cut off wavelength must be $(167-84)$ pm $= 83$ pm.

Now $eV = \dfrac{hc}{\lambda_c} \Rightarrow eV = \dfrac{12420\,eV\,A^\circ}{0.83\,A^\circ} \Rightarrow V = 15$ kVolts

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Example: Irodov: 6.137:

At a certain voltage applied to an X-ray tube with aluminium anticathode the short -wave cut-off wavelength of the continuous X-ray spectrum is equal to 0.50 nm. Will the K series of the characteristic spectrum whose excitation potential is equal to 1.56 kV be also observed in this case? [Here excitation potential means required potential to knock out K electrons i.e $K \to \infty$]

Solution: $eV = \frac{hc}{\lambda_{cut}} = \frac{1242\ eV\ nm}{0.5\ nm} \approx 2.5\ keV$, thus

accelerating voltage is more than voltage required to produce Kα excitation $\therefore$ Yes K series emission will be observed.

Example: Irodov: 6.138:

When the voltage applied to an X-ray tube increased from $V_1 = 10$ kV to $V_2 = 20$ kV, the wavelength interval between the k_α line and the short-wave cut-off of the continuous X-ray spectrum increases by a factor $n = 3.0$. Find the atomic number of the element of which the tube's anticathode is made.

Solution: Cut off wavelength in first case, $\lambda_1 = \frac{hc}{eV} = \frac{1242\ eV\ nm}{e \times 10\ kV}$

$\Rightarrow \lambda_1 \approx 124$ pm similarly in second case, $\lambda_2 = 62$ pm.

Now let λ be Kα wavelength then $\frac{\lambda - 62}{\lambda - 124} = 3 \Rightarrow \lambda = 155$ pm.

Now $\lambda = \frac{4\lambda_0}{3(z-1)^2} \Rightarrow z = \sqrt{\frac{4\lambda_0}{3\lambda}} + 1 = \sqrt{\frac{4 \times 911\text{Å}}{3 \times 1.55\text{Å}}} + 1 \approx 29$

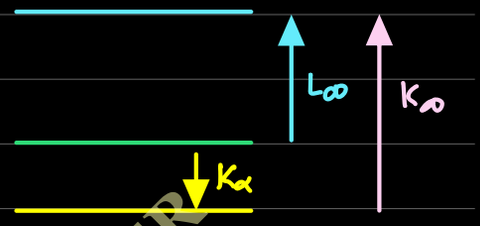
Thus required atomic number = 29 i.e Cu.

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EXAMPLE : IRODOV : 6.139 :

What metal has in its absorption spectrum the difference between angular frequencies of X-ray K and L absorption edges equal to $\Delta\omega = 6.85 \times 10^{18}\ \text{rad s}^{-1}$? (Here absorption edge refers to the transition $K \to \infty$ and $L \to \infty$).

Solution: $h\nu_{K\alpha} = h\nu_{K\infty} - h\nu_{L\infty}$



(This can be readily understood in terms of energy level diagram in the figure).

Thus $\Delta\nu = \nu_{K\alpha} = \nu_{K\infty} - \nu_{L\infty} = A(Z-1)^2$

where $A = 3Rc/4$, thus, $Z = \sqrt{\frac{\Delta\nu}{A}} + 1$

$$\Rightarrow Z = \sqrt{\frac{4\Delta\omega \times h}{3Rc \times h \times 2\pi}} + 1$$

Now $Rhc = 13.6\ \text{eV}$, $h = 4.14 \times 10^{-15}\ \text{eV sec}$

Thus, $Z = \sqrt{\frac{4 \times 6.85 \times 10^{18}\ \text{Hz} \times 4.14 \times 10^{-15}\ \text{eV sec}}{3 \times 13.6\ \text{eV} \times 2\pi}} + 1 \approx 22$

Thus required metal is $Z = 22$ (Titanium).

→ Irodov language is ambiguous. It says frequency $\Delta\omega = 6.85 \times 10^{18}\ \text{s}^{-1}$ when infact it means angular frequency, hence the factor of 2π in the denominator in the square root.

EXAMPLE : IRODOV : 6.141 :

Find the binding energy of an K electron in titanium if the wavelength difference between the first line of the K series and its short-wave cut-off is $\Delta\lambda = 26$ pm. ($Z = 22$)

Solution: Binding energy in K means $E_{K\infty}$ short wave cut-off from K shell corresponds to $E_{K\infty}$, thus $\lambda_{K\alpha} - \lambda_{K\infty} = 26\ \text{pm}$ according to the question. Now $\lambda_{K\alpha} = \frac{4\lambda_0}{3(Z-1)^2} = \frac{4 \times 911\ \text{Å}}{3 \times 21^2} \approx 275\ \text{pm}$.

Thus $\lambda_{K\infty} = \lambda_{K\alpha} - 26\ \text{pm} = 275 - 26 = 249\ \text{pm}$.

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Now $E_{K\alpha} = \frac{hc}{\lambda_{K\alpha}} = \frac{12420}{2.49}$ eV $\approx$ 5 keV Ans.

EXAMPLE: IRODOV: 6.142:

Find the kinetic energy and the velocity of the photoelectrons liberated by K_α radiation of zinc from the K shell of iron whose K band absorption edge wavelength is $\lambda_K = 174$ pm. $Z_{Zn} = 30$, $Z_{Fe} = 26$

Solution: Here Energy of Photoelectrons must be $(E_{K\alpha})_{Zn} - (E_{K\alpha})_{Fe}$

Now $(E_{K\alpha})_{Zn} = 13.6 \times (30-1)^2 \times \frac{3}{4}$ eV $= 8.57$ keV

$(E_{K\alpha})_{Fe} = \frac{1242 \text{ eV nm}}{174 \text{ pm}} = 7.14$ keV Thus required energy of photoelectrons, $KE = (8.57 - 7.14)$ keV ≈ 1.43 keV Ans.

$m_e = 511 \text{ keV}/c^2$

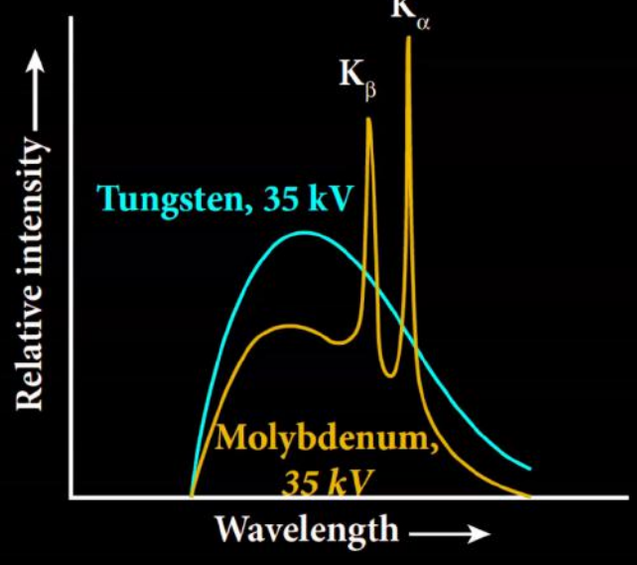
EXAMPLE: Glancing angle for monochromatic x-rays from a crystal whose atomic spacing is 2.5 Å is 15°. Calculate the wavelength of the x-rays. (Assume first order maximum).

Solution: By Bragg's law, $2d \sin\theta = n\lambda$, putting $n = 1$,

$\lambda = 2 \times 2.5 \text{ Å} \times \sin 15° = 1.29 \text{ Å} = 129$ pm Ans.

EXAMPLE:

The adjacent graph shows the x-ray spectrum for a tungsten and a molybdenum targets for same accelerating voltages. Why are K_α and K_β lines missing from the tungsten graph?



Solution: To get K series lines, electron energy must be greater than K_α, thus from graph we can conclude that K_α

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excitating potential for tungsten must be more than 35 kilo Volt.

Uses Of X-Rays:

1. **Medical Diagnosis:** All of us are familiar with x-ray radiographs for determining fractures / internal bone structures. For medical radiology, usually soft x-rays (Typically less than 5keV, x-rays of energy above 10 keV are called hard x-rays between 5keV to 10keV are "moderate").

2. **Medical Therapy:** X-Rays are high energy radiation and thus can be used for killing diseased tissues. They are used for curing skin diseases or malignant tumours. (Though it must be done carefully, x-rays can as well kill healthy tissues upon over exposure and therefore administered in controlled doses. The radiation dose is typically measured in milli Sieverts or Joules/Kg or rads and Grays).

3. **Industry:** X-Rays are used for detecting flaws in welded joints, motor tyres, tennis balls and wood. At security scanners, X-Rays are used for detection of contraband/illegal items.

4. **Scientific Research:** X-Ray Diffraction is used to study the structure of crystalline materials i.e arrangement of atoms and molecules in various crystals.

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Example: JEE 2020 P2 Q10: (Wrong Key By JIT):

In an X-ray tube, electrons emitted from a filament (cathode) carrying current I hit a target (anode) at a distance d from the cathode. The target is kept at a potential V higher than the cathode resulting in emission of continuous and characteristic X-rays. If the filament current I is decreased to $\frac{I}{2}$, the potential difference V is increased to $2V$, and the separation distance d is reduced to $\frac{d}{2}$, then

(A) the cut-off wavelength will reduce to half, and the wavelengths of the characteristic X-rays will remain the same

(B) the cut-off wavelength as well as the wavelengths of the characteristic X-rays will remain the same

(C) the cut-off wavelength will reduce to half, and the intensities of all the X-rays will decrease

(D) the cut-off wavelength will become two times larger, and the intensity of all the X-rays will decrease

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Solution: Since potential difference is doubled therefore electron energies are also double and so also cut off photon energies and frequencies. Thus cut-off wavelength is halved. (option A is correct).

Since the second trial has smaller cut off frequency it will be higher than the first case graph near the cut off of the first case. Thus C is wrong

But JEE gave C also as the correct key

